

Project 2

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Dataset:

☆ Chinese Yuan Renminbi to U.S. Dollar Spot Exchange Rate (DEXCHUS)

Observations 

2025-11-28: 7.0751

Updated: Dec 1, 2025 3:18 PM CST

Next Release Date: Dec 8, 2025

Units:

Chinese Yuan Renminbi to One U.S. Dollar,
Not Seasonally Adjusted

Frequency:

Daily

1Y

5Y

10Y

Max

2020-11-28

to

2025-11-28

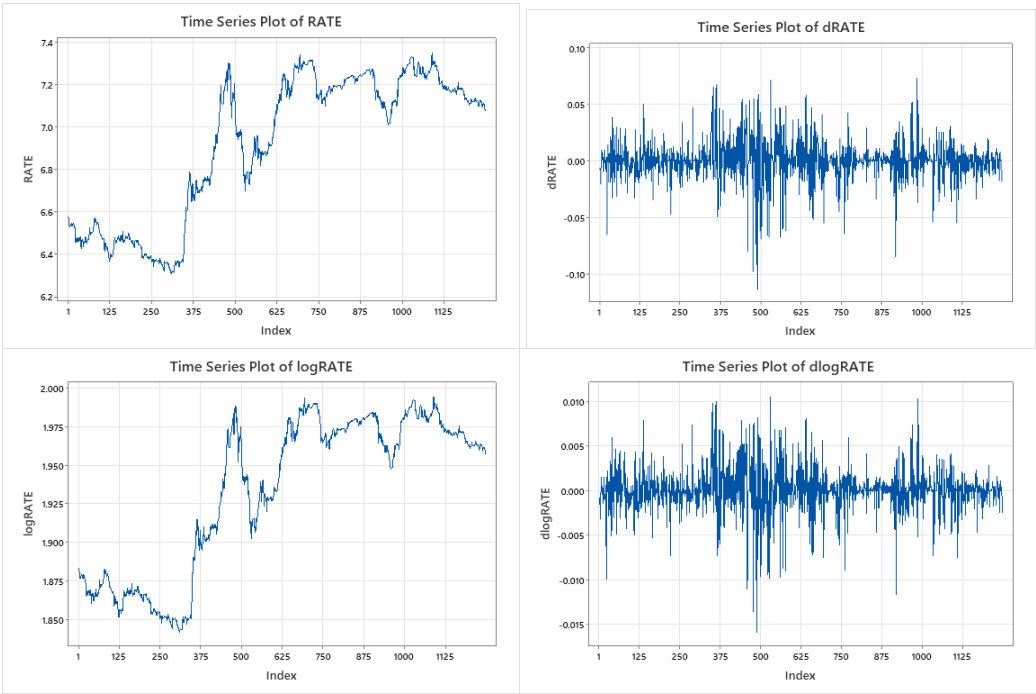
Source: <https://fred.stlouisfed.org/series/DEXCHUS>

The data set is the daily Chinese Yuan Renminbi to U.S. Dollar spot exchange rate from FRED (series code DEXCHUS). Each observation is the number of yuan required to buy one U.S. dollar on a given trading day, so the units are “CNY per USD,” not seasonally adjusted. The series is recorded at a daily frequency and reflects movements in the offshore/onshore CNY–USD market driven by macroeconomic news, monetary policy, and global risk sentiment. For the project I use the first 1248 daily observations as the estimation sample and keep the 1249th observation (the most recent day) as a hold-out point to evaluate the ARMA–ARCH forecast interval.

observation_date	RATE	logRATE	dlogRATE
2025-11-28	7.0751	1.95658	*

Part 1

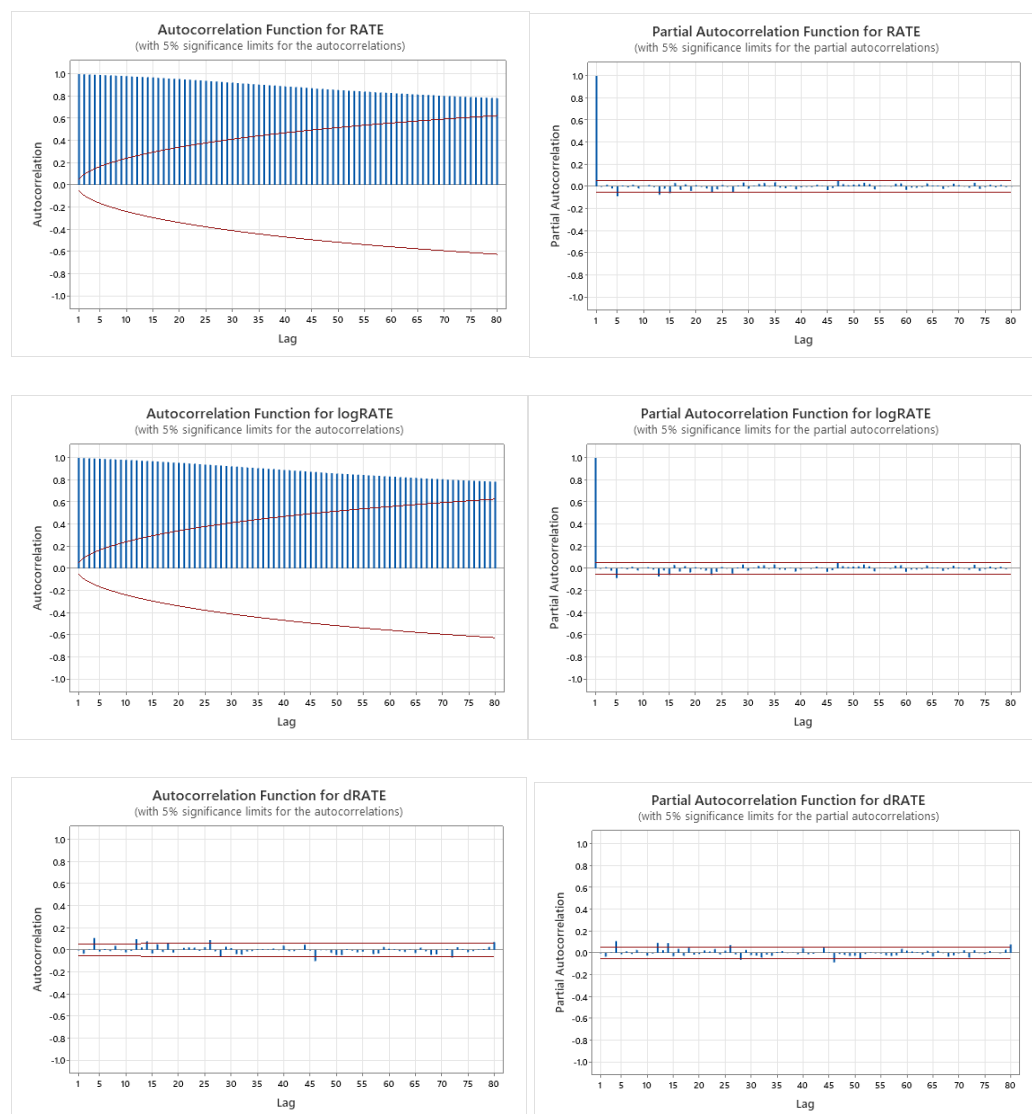
Level and log-level plots

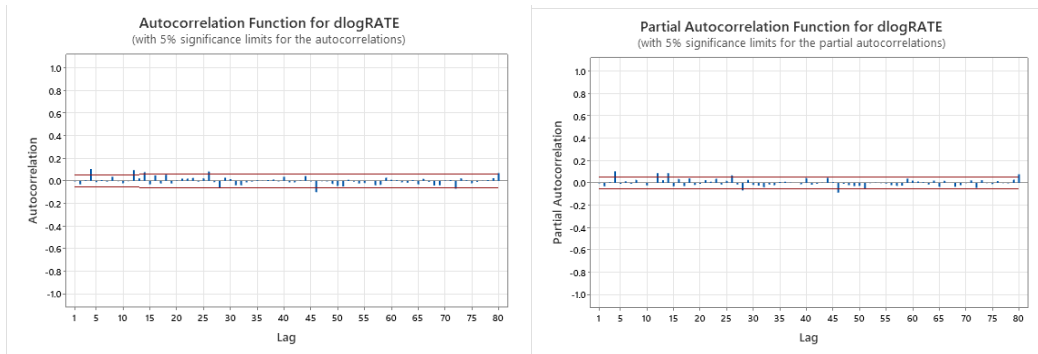


The plot of dlogRATE shows clear **volatility clustering**: intervals of relatively calm fluctuations are interrupted by episodes with many large spikes. This suggests that the **conditional variance** of returns changes over time, rather than being constant.

The series is strictly positive, and in exchange-rate applications it is more natural to model relative (percentage) changes. The time plots of RATE and logRATE look very similar, so a log transformation is not essential for variance stabilization, but we work with logRATE so that first differences of logs can be interpreted as **approximate percentage changes**.

ACF and PACF





From the plots:

- **logRATE**: the time-series plot shows persistent level shifts and long swings; the ACF of logRATE is very large at lag 1 and then decays only slowly. This is typical of a **non-stationary** series with a unit root.
- **dlogRATE**: the differenced series fluctuates around a constant mean with roughly constant variance, and its ACF and PACF are essentially within the 5% bounds at all lags, indicating an approximately **white-noise** process.

So the original series does **not** appear stationary, but its first difference does. This suggests $d = 1$. Because the ACF and PACF of $\Delta \log \text{RATE}_t$ show no strong spikes at low lags, a natural model is

$$\text{ARIMA}(p, d, q) = \text{ARIMA}(0, 1, 0), \text{ i.e. a random walk for logRATE (without a constant).}$$

Part 2

I fit $\text{ARIMA}(p, 1, q)$ to $y_t = \log(\text{RATE})$ for $p, q \in \{0, 1, 2\}$, both with and without a constant. The AICc minimum (-11310.2) occurs at $\text{ARIMA}(0, 1, 0)$ without constant, i.e. a random walk.

Forecast with Best ARIMA Model for logRATE with Constant

* WARNING * Inestimable ARIMA(p, d, q) models that include a constant term:
(1, 1, 1), (1, 1, 2), (2, 1, 1), (2, 1, 2)

Method

Criterion for best model Minimum AICc
Rows used 1248
Rows unused 0

Model Selection

Model (d = 1)	LogLikelihood	AICc	AIC	BIC
p = 0, q = 0*	5656.40	-11308.8	-11308.8	-11298.5
p = 0, q = 1	5656.41	-11306.8	-11306.8	-11291.4
p = 1, q = 0	5656.41	-11306.8	-11306.8	-11291.4
p = 1, q = 1	5656.13	-11306.2	-11306.3	-11290.9
p = 2, q = 0	5657.01	-11306.0	-11306.0	-11285.5
p = 0, q = 2	5656.91	-11305.8	-11305.8	-11285.3
p = 1, q = 2	5656.57	-11305.1	-11305.1	-11284.6

* Best model with minimum AICc.

Forecast with Best ARIMA Model for logRATE without Constant

* WARNING * Inestimable ARIMA(p, d, q) models that do not include a constant term:
(2, 1, 1), (2, 1, 2)

Method

Criterion for best model Minimum AICc
Rows used 1248
Rows unused 0

Model Selection

Model (d = 1)	LogLikelihood	AICc	AIC	BIC
p = 0, q = 0*	5656.08	-11310.2	-11310.2	-11305.0
p = 0, q = 1	5656.09	-11308.2	-11308.2	-11297.9
p = 1, q = 0	5656.09	-11308.2	-11308.2	-11297.9
p = 2, q = 0	5656.67	-11307.3	-11307.3	-11292.0
p = 0, q = 2	5656.57	-11307.1	-11307.1	-11291.8
p = 1, q = 1	5656.13	-11306.2	-11306.3	-11290.9
p = 1, q = 2	5656.57	-11305.1	-11305.1	-11284.6

* Best model with minimum AICc.

Selected Model and Full Equation

Final Estimates of Parameters

Random walk model without constant

Model Summary

DF	SS	MS	MSD	AICc	AIC	BIC
1246	0.0083819	0.0000067	0.0000067	-11310.2	-11310.2	-11305.0

MS = variance of the white noise series

Let $Y_t = \text{logRATE}_t$.

$$\nabla Y_t = Y_t - Y_{t-1} = a_t, \quad a_t \sim \text{WN}(0, \sigma_a^2),$$

so in level form:

$$Y_t = Y_{t-1} + a_t \Leftrightarrow \text{logRATE}_t = \text{logRATE}_{t-1} + a_t, a_t \sim \text{WN}(0, \sigma_a^2)$$

(with $\sigma_a^2 \approx 0.0000067$ from the Model Summary).

One-step-ahead ARIMA forecast is 1.95668 and 95% forecast interval is [1.95160, 1.96177]

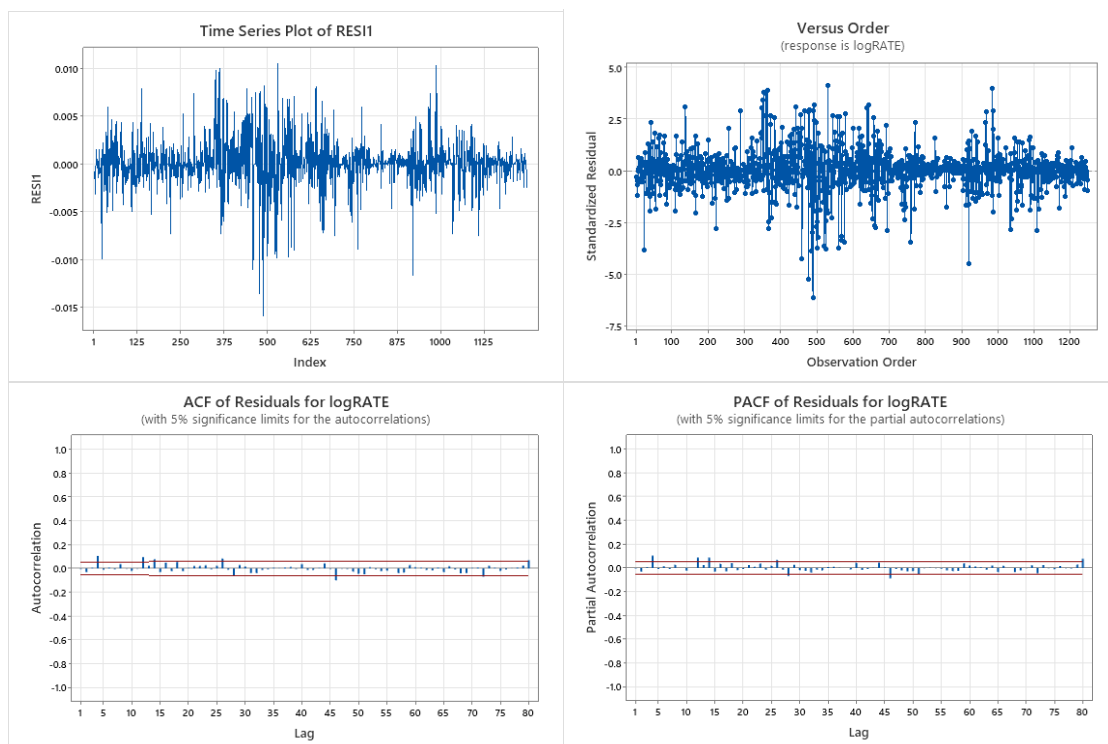
Forecasts from Time Period 1248

Time Period	Forecast	SE Forecast	95% Limits		Actual
			Lower	Upper	
1249	1.95668	0.0025937	1.95160	1.96177	

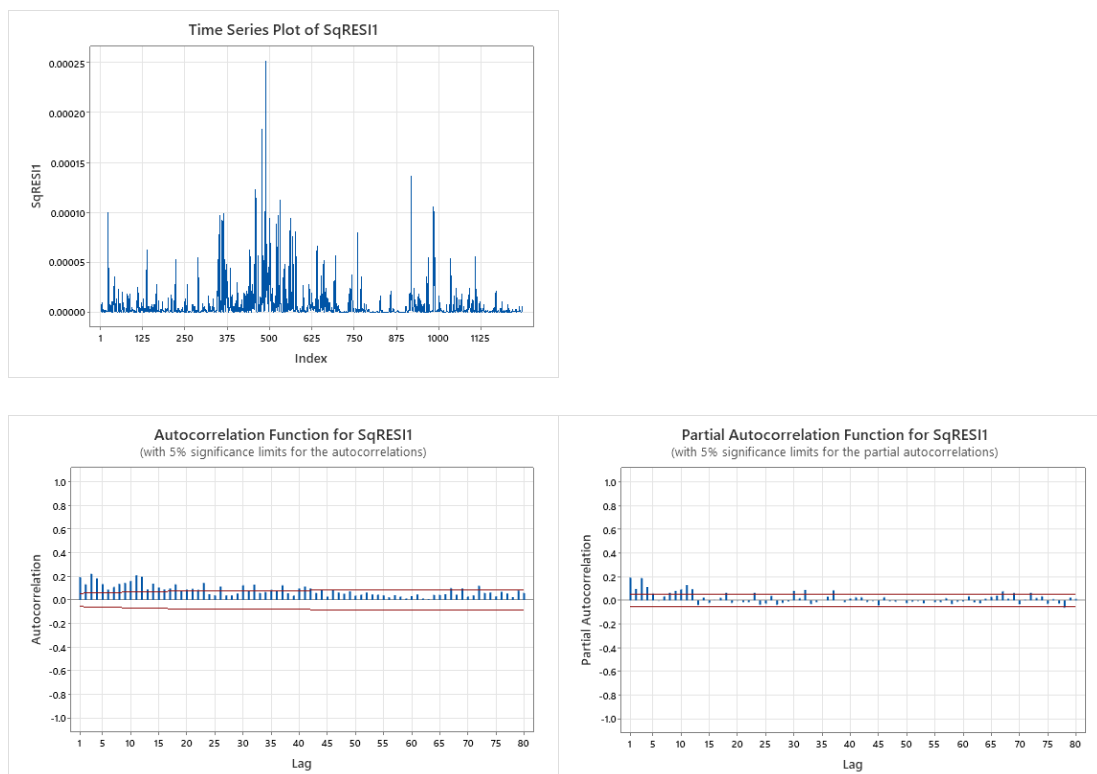
Part 3

Residuals Plots:

The time series plot and the residual-versus-order plot show residuals fluctuating around zero with roughly constant spread. The ACF and PACF of the ARIMA residuals are essentially flat, with all sample autocorrelations inside the 5% significance bands at the lags considered. This indicates that the residuals are approximately uncorrelated and that the ARIMA(0,1,0) model has adequately captured the mean dynamics of logRATE.



Squared residuals plots:



(Ljung-Box) Chi-Square Statistic

Modified Box-Pierce (Ljung-Box) Chi-Square Statistic

Lag	12	24	36	48
Chi-Square	29.01	48.28	68.35	85.88
DF	12	24	36	48
P-Value	0.004	0.002	0.001	0.001

The picture changes when we examine the squared residuals $\hat{\varepsilon}_t^2$. The time-series plot of $\hat{\varepsilon}_t^2$ displays clear volatility clustering: long stretches of relatively low variance are interrupted by episodes of markedly higher variance. The ACF of $\hat{\varepsilon}_t^2$ shows several sizeable positive autocorrelations at low lags that decay only slowly, and the PACF also has distinct spikes at small lags. If the residuals were independent and identically distributed with constant variance, then functions such as $\hat{\varepsilon}_t^2$ would be approximately uncorrelated and their ACF/PACF would resemble white noise.

The strong dependence in the squared residuals therefore indicates conditional heteroskedasticity: the conditional variance of $\hat{\varepsilon}_t$ depends on past information. This is confirmed by the Ljung–Box statistics for $\hat{\varepsilon}_t^2$, whose p-values at lags 12, 24, 36 and 48 (0.004, 0.002, 0.001, 0.001) are all highly significant, so we strongly reject the null of no serial correlation in the squared residuals.

ARIMA(0,1,0) thus provides an adequate model for the conditional mean (according to AICc and the residual ACF/PACF), but the variance is time-varying. Therefore it is natural to extend the model to an ARIMA(0,1,0)–GARCH(1,1) specification for the conditional variance.

Part 4

ARCH(0): log likelihood by hand

Formula for ARCH(q) AICc (N = effective sample size) is $\text{AICc} = -2\log L + \frac{2(q+1)N}{N-q-2}$.

For $q = 0$ the log-likelihood is

```
> N <- length(x)
> ll0 <- -0.5 * N * (1 + log(2 * pi * mean(x^2)))
> aicc0 <- -2 * ll0 + 2 * (0 + 1) * N / (N - 0 - 2)
```

ARCH(q), $q = 1, \dots, 10$: use garch() and logLik()

```
> results <- data.frame(q = 0:10, logLik = NA, AICc = NA)
>
> ## fill in q = 0
> results$logLik[results$q == 0] <- ll0
> results$AICc[results$q == 0] <- aicc0
>
> ## q = 1, ..., 10
> for (q in 1:10) {
+   fit <- garch(x, c(0, q))          # ARCH(q)
+   ll <- as.numeric(logLik(fit))      # log likelihood
+   aicc <- -2 * ll + 2 * (q + 1) * N / (N - q - 2)
+
+   results$logLik[results$q == q] <- ll
+   results$AICc[results$q == q] <- aicc
+ }
```

Result Table

```
> results
  q  logLik  AICc
1  0 5656.079 -11310.16
2  1 5714.046 -11424.08
3  2 5728.569 -11451.12
4  3 5763.645 -11519.26
5  4 5775.796 -11541.54
6  5 5776.485 -11540.90
7  6 5773.437 -11532.78
8  7 5773.969 -11531.82
9  8 5773.023 -11527.90
10 9 5771.851 -11523.52
11 10 5767.431 -11512.65
```

The smallest AICc among the ARCH models occurs at $q=4$.

Also fit a GARCH(1,1) model and compare their AICc:

```
> aicc_table
  model  logLik  AICc
1  ARCH(4) 5775.796 -11541.54
2  GARCH(1,1) 5821.308 -11636.60
```

GARCH(1,1) has a smaller AICc and therefore is preferred.

The $\log\text{Lik}(\text{GARCH}(1,1)) = 5821.308$ and $\text{AICc} = -11636.60$. Below is the summary of the GARCH(1,1) model.

```
> summary(fit_g11)
```

Call:

```
garch(x = x, order = c(1, 1))
```

Model:

GARCH(1,1)

Residuals:

Min	1Q	Median	3Q	Max
-6.178906	-0.416606	0.005727	0.462685	5.028608

Coefficient(s):

	Estimate	Std. Error	t value	Pr(> t)
a0	1.286e-07	1.980e-08	6.494	8.33e-11 ***
a1	8.834e-02	7.408e-03	11.925	< 2e-16 ***
b1	8.942e-01	8.204e-03	109.002	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Diagnostic Tests:

Jarque Bera Test

data: Residuals

X-squared = 1270.4, df = 2, p-value < 2.2e-16

Box-Ljung test

data: Squared.Residuals

X-squared = 0.12652, df = 1, p-value = 0.7221

The statistical significance of the parameter values of GARCH(1,1) model:

Let e_t be the ARIMA residual and $h_t = \text{Var}(e_t | \mathcal{F}_{t-1})$. From the summary:

- $a_0 = 1.286 \times 10^{-7}$
- $a_1 = 8.834 \times 10^{-2}$
- $b_1 = 8.942 \times 10^{-1}$

So the fitted GARCH(1,1) is $e_t | \mathcal{F}_{t-1} \sim N(0, h_t)$,
 $h_t = 1.286 \times 10^{-7} + 0.08834 e_{t-1}^2 + 0.8942 h_{t-1}$.

All three coefficients have huge t-values and p-values $< 2e-16$, so they are highly significant. Also note $a_1 + b_1 \approx 0.982 < 1$, so the model is (covariance-)stationary with strong volatility persistence.

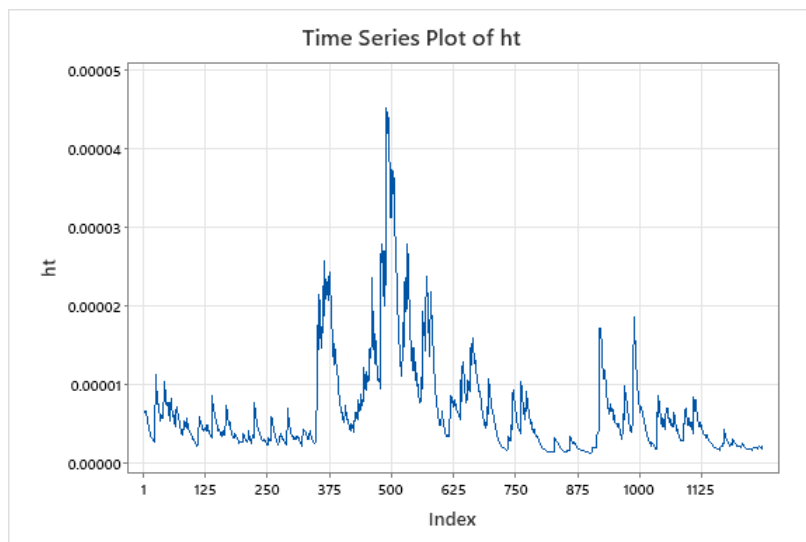
Unconditional (marginal) variance of the shocks in this model:

```
> # Unconditional variance and standard deviation of shocks e_t
> var_uncond <- omega / (1 - alpha1 - beta1)
> sd_uncond  <- sqrt(var_uncond)
> var_uncond
a0
7.368442e-06
> sd_uncond
a0
0.002714487
```

The unconditional (marginal) variance of the shocks in the fitted GARCH(1,1) model is 7.37×10^{-6} , corresponding to an unconditional standard deviation of about 0.0027.

Part 5 & 6

Time Series Plot of the conditional variances (h_t) of GARCH(1,1) model:



There are pronounced bursts of high volatility around indices roughly 350–650, with a particularly sharp peak near index ≈ 500 where h_t reaches its maximum. Smaller volatility clusters appear earlier (around indices 50–150) and later (around indices 800–950), where h_t rises noticeably above its baseline level. These volatility bursts coincide with the large upward and downward moves visible in the logRATE series, confirming that the GARCH model is capturing periods of market turbulence.

A 95% one step ahead forecast interval for the logRATE, based on the ARIMA-ARCH model is

[1.953675, 1.959685] and the 5th percentile is **1.954158**.

```
> h_next      <- omega + alpha1 * e_last2 + beta1 * h_last
> sqrt_h_next <- sqrt(h_next)
>
> h_next
[1] 2.35055e-06
> sqrt_h_next
[1] 0.00153315

> # ARIMA forecast for logRATE_{n+1} from Minitab (Problem 2)
> f_n1 <- 1.95668
>
> # 95% one-step-ahead PI for logRATE_{n+1}
> lower_95 <- f_n1 - 1.96 * sqrt_h_next
> upper_95 <- f_n1 + 1.96 * sqrt_h_next
> c(lower_95, upper_95)
[1] 1.953675 1.959685
>
> # 5th percentile of conditional distribution N(f_n1, h_{n+1})
> q05 <- f_n1 + qnorm(0.05) * sqrt_h_next
> q05
[1] 1.954158
```

Forecasts from Time Period 1248

		95% Limits		
Time Period	Forecast	SE Forecast	Lower	Upper
1249	1.95668	0.0025937	1.95160	1.96177

The interval based on the ARIMA only model: [1.95160, 1.96177]

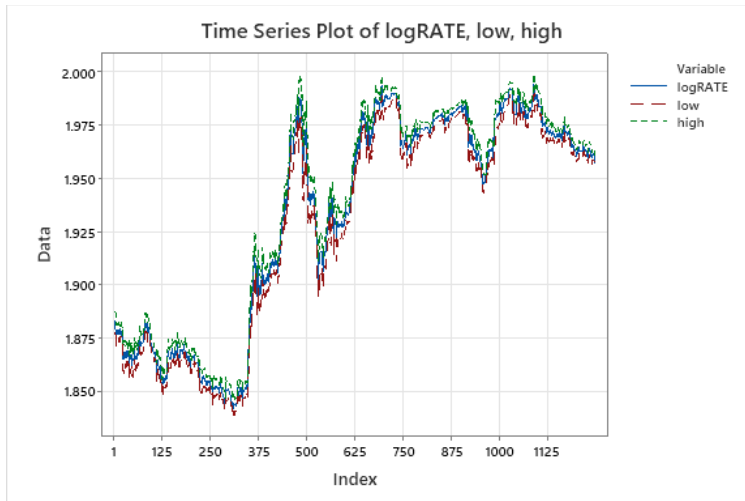
Comparison on the intervals:

ARIMA-only interval: [1.95160, 1.96177]

ARIMA-GARCH(1,1) interval: [1.95368, 1.95969]

The actual value $\log RATE_{1249} = 1.95658$ lies well inside both intervals, essentially near their centers, so both models provide reasonable coverage for this date. However, the ARIMA–GARCH interval is noticeably narrower, because the GARCH component allows the conditional variance at the forecast origin to be relatively low. In contrast, the ARIMA-only interval is based on a constant innovation variance and therefore does not adjust the width of the forecast interval to periods of calm or turbulent volatility.

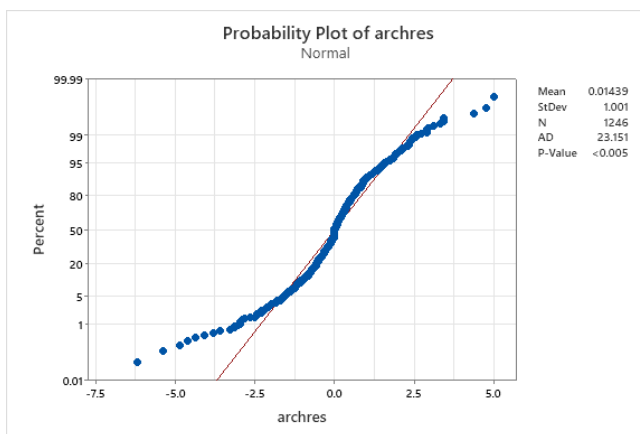
Part 7



The plot shows that the forecast intervals adapt well to the behavior of the log exchange rate. During volatile periods, the bands widen to accommodate the larger price movements, while in more tranquil periods they become noticeably tighter. This time-varying behavior allows the intervals to capture the vast majority of observations, including many that occur during turbulent market phases where a constant-width ARIMA interval would likely fail. From a practical perspective, these conditional-volatility intervals are much more informative than static ARIMA intervals, because they provide a day-by-day assessment of likely ranges for the next observation that reflects the most recent level of market volatility.

Part 8

Normal probability plot of archres:



After standardizing by $\sqrt{h_t}$, the residuals have mean close to 0 and variance close to 1, but the normal probability plot shows substantial departures from the straight line in both tails and the Anderson–Darling test strongly rejects normality ($p < 0.005$). This indicates that, although the ARIMA–GARCH model captures the time-varying volatility and removes serial dependence in the residuals, it **does not fully account for the leptokurtosis in the data**: the standardized residuals still have noticeably heavier tails than a normal distribution. As a consequence,

prediction intervals that rely on the normal assumption tend to be somewhat too narrow, with an empirical failure rate greater than the nominal 5%.

Part 9

failures
81

There are 1246 standardized residuals in total. With 81 failures, failure rate $= \frac{81}{1246} \approx 0.065 \approx 6.5\%$.

So the nominal 95% prediction intervals failed to cover the next day's log exchange rate about **6.5% of the time**.

Part 10

The last data point, $\log RATE_{1249} = 1.95658$, lies well inside both one-step-ahead forecast intervals: it is essentially at the center of both the ARIMA-only interval $[1.95160, 1.96177]$ and the ARIMA–GARCH interval $[1.95368, 1.95969]$. For this particular forecast date, both intervals therefore appear “about right”. Looking at the sample as a whole, however, the conclusions differ. The ARIMA-only intervals are relatively wide and tend to miss less often than the nominal 5%, so they are slightly conservative and ignore volatility clustering. The ARIMA(0,1,0)–GARCH(1,1) model substantially improves the description of time-varying volatility and produces tighter, state-dependent intervals that adapt to current market conditions. Nevertheless, the standardized residuals from the ARIMA–GARCH model remain heavy-tailed, so normal-based 95% prediction intervals are still somewhat too narrow, with an empirical failure rate (6.5%) slightly above 5%. Thus the ARIMA–GARCH model is an improvement over ARIMA-only for modeling conditional variance, but it does not fully capture tail risk.

Overall, an ARIMA(0,1,0) model with GARCH(1,1) innovations provides a much better description of the exchange rate dynamics than a pure ARIMA model: it captures volatility clustering and produces informative, time-varying prediction intervals.